



The Intuition of Maximum Likelihood Transcript

Hello everyone, welcome to lesson 1 video 1. In this lesson, we will develop the intuition for Maximum Likelihood Estimation. Rather than guessing parameters, Maximum Likelihood Estimation asks us which parameter values make the most observed data most probable. Let's learn about Maximum Likelihood Estimation or MLE.

This is one of the methods which we used for finding the best explanation for our data. In this method, we will try using our understanding of how do we explain the data that we have got using the best likelihood or the best likely parameter which can be fitted given the distribution and so on. But you will get a good understanding of how can you easily use this technique to get to the best parameter which explains the data.

With that, let's begin. Let's understand all of these things with intuition first. Let's say you have a coin and you have been given that coin by a stranger or someone you know.

Now, you don't know if it is a fair coin or not. You do not know about it. A fair coin is a coin wherein the likelihood of or the chances of getting a head or tail both are 0.5. So, if it is greater than 0.5, it is biased in a new way.

What if you flip that coin 10 number of times and you get this data? Basically what I mean by this data is you have 8 heads and 2 tails. Now given that you have got or you have flipped the coin 10 number of times and you have got 8 heads and 2 tails, what is the best guess of the probability of getting head with this coin? Or in other words, can you say that given the kind of data that you have got, can you say that the coin is a biased coin and it is biased towards heads? So, that's what we are going to understand. That's how we use maximum likelihood estimate.

Generally, what we do is we assume a distribution and then we tell about probability or the chances that are likely to happen. But herein, we are first looking at the data and then understanding from what the data might have been coming from or from what distribution the data might be coming from. That's the difference between what you call as probability and the likelihood.

Now thinking of maximum likelihood estimator, how do you actually get to the value of p or let's say for example, if you have in the previous case, we got this value of p which was let's say probability of getting a head. Now given that, you don't know about whether it's a fair coin or it's not a fair coin, but you only know about this data. Now what is the best guess for your true probability is what we are trying to understand whether the probability of getting heads is 0.2 or 0.5 or 0.8. If it is probability of getting head is 0.8, it directly means that the coin is unfair and that's what I told you initially.



I told you that given this data, can you tell whether the coin is unfair towards head right now because you have got 8 heads out of 10. So given all of that, what if the probability is equal to 0.5? You do not know about that, you do not know about the value of p given the fair coin or whether it is a fair coin or it is not a fair coin and you do not know whether the value is going to be 0.1. If the value is going to be 0.1, it is likely a very biased coin, it is biased towards tails. The value is going to be 0.8, it is going to be biased towards head and this is what we should be looking for.

So if we get a value probability of getting head p is equals to let's say p is the probability of getting head. If we are getting probability of getting head as 0.8, it means that if you toss that same coin 10 number of times, you will get exactly same observations or exactly same results that we discussed previously. So you will get 8 heads and 2 tails.

So given the data we observed, what is the value of the parameter? Here in the parameter is the probability, what is the value of the parameter given that data is most likely to be observed. Given that data, you have already known the data, what is the value of the parameter would make this observed data more likely to be happened? Let's say what is the value of the parameter which gives more chances to the kind of data that you have got or which gives more chances of observing the same kind of data that you have got is what is maximum likelihood estimation and everything in maximum likelihood estimation revolves around knowing the parameter value. Here in the parameter value is probability of getting a head.

So let's say that we now formalise the entire definition. So basically MLE or maximum likelihood estimation for the given parameter is a value of that parameter that maximises the probability of observing the data you actually collected. You already have the data, now we are looking for the value of the parameter which maximises the probability.

It doesn't give you the probability directly, it says that value of the parameter which maximises the probability, that's why it is called as maximum likelihood. Given the data, what is the value of the parameter which maximises the probability of observing the data you have collected? So in your coin flip example, your intuition was the probability is equal to 0.8. This is the most possible value, right? Because you have already got the data as 8 heads and 2 tails, hence 0.8 or 0.9 would make sense rather than 0.1. In summary, maximum likelihood estimation methods is all about choosing the parameter that makes sense to the data. This simple but powerful idea underpins much more deeper insight into the method which we are learning.

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